

MoCoO: Momentum Contrast ODE-Regularized VAE for Single-Cell Trajectory Inference and Representation Learning

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Abstract—Learning biologically faithful low-dimensional representations from single-cell RNA sequencing (scRNA-seq) data requires models that simultaneously preserve discrete cell-type identity and continuous developmental dynamics. We present MoCoO, a modular framework that integrates a Variational Autoencoder (VAE), Neural Ordinary Differential Equations (Neural ODE), and Momentum Contrast (MoCo) with learnable prototypes for robust trajectory inference and representation learning. Through a systematic six-configuration ablation on the IRALL hematopoiesis dataset (3,000 cells, 200 epochs) with a KL weight sweep ($\beta \in \{1.0, 0.1, 0.01\}$), we investigate the standalone power and synergistic interactions of each component. Key findings: (1) ODE is the dominant component—it consistently reduces Davies–Bouldin index (DB; lower is better) by 0.28–0.62 across all β , winning 13 of 27 metric– β combinations; (2) MoCo has beta-dependent standalone power—contrastive learning improves ARI by +0.039 at $\beta = 0.01$ and DREX by +0.006 at $\beta \leq 0.1$, but is ineffective at strong KL; (3) ODE and MoCo exhibit regime-dependent synergy—the interaction term for DB reaches -0.129 at $\beta = 0.01$ (super-additive compactness), with DRE (+0.031) and DREX (+0.030) synergies confirming improved embedding fidelity at low β ; (4) the Full model achieves the best distance preservation across configurations, with the highest DRE at $\beta = 0.01$ (0.518) and strongest pairwise distance correlations (DREX Spearman up to 0.72). Subcategory analysis reveals that while VAE+ODE dominates aggregate clustering metrics, the Full model uniquely excels at geometric fidelity of the learned representation. Pseudotime–marker correlations are biologically significant across five systems: *Hbb-bs* ($\rho = 0.28$) in IRALL, *Slc1a3* ($\rho = 0.54$) in dentate, *Ins2* ($\rho = 0.46$) in endo, *Epor* ($\rho = 0.35$) in paul, and *MKI67* ($\rho = 0.49$) in spinoids. We publicly release the MoCoO Python package and benchmark suite.

Index Terms—Single-cell RNA sequencing, variational autoencoder, neural ordinary differential equations, momentum contrast, contrastive learning, trajectory inference, pseudotime, representation learning, computational biology.

I. INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) has transformed our understanding of cellular heterogeneity by enabling transcriptome-wide profiling at single-cell resolution [1]. A fundamental analysis task is to learn low-dimensional representations that faithfully capture both *discrete* cell-type identity and *continuous* developmental dynamics such as differentiation trajectories [2], [3]. These representations underpin downstream tasks including clustering, pseudotime inference,

RNA velocity estimation, and differential expression analysis [4].

Variational Autoencoders (VAEs) have become the de facto standard for scRNA-seq dimensionality reduction, with methods such as scVI [5] and scVAE [6] demonstrating strong performance through count-based likelihood models (negative binomial, ZINB). However, standard VAE latent spaces tend to be over-smoothed, conflating nearby cell states and producing suboptimal trajectory structures. Two complementary strategies have emerged to address this limitation.

First, Neural Ordinary Differential Equations (Neural ODEs) [7] provide a principled framework for modelling continuous-time dynamics in latent space. By parameterising the derivative of the latent state with a neural network and integrating forward through an ODE solver, these models can learn smooth developmental trajectories without discrete time-step assumptions. Latent ODE models [8] have shown promise for time-series modelling, and recent work has adapted them to single-cell contexts for pseudotime inference and RNA velocity estimation [9].

Second, contrastive learning—particularly Momentum Contrast (MoCo) [10]—has proven effective at learning representations that are locally smooth yet globally discriminative. MoCo maintains a momentum-updated encoder and a large memory queue of negative keys, enabling scalable contrastive learning that is decoupled from batch size. Extensions such as prototype contrastive learning [11] further impose cluster structure by aligning representations with learnable prototype vectors.

Despite these individual advances, no existing method unifies VAE-based reconstruction, Neural ODE dynamics, and momentum contrastive learning within a single coherent framework. In this work, we propose MoCoO (**M**omentum **C**ontrast **O**DE-regularized VAE), a modular architecture that combines all three paradigms:

- 1) **VAE with flexible count-based likelihoods** (MSE, NB, ZINB, Poisson, ZIP) provides a probabilistic latent space and handles the zero-inflation and overdispersion characteristic of scRNA-seq count data.
- 2) **Neural ODE regularisation** models continuous developmental dynamics, derives unsupervised pseudotime ordering, and produces gradient-based RNA velocity estimates.
- 3) **Momentum Contrast with prototype heads** imposes instance-level contrastive regularisation via a momentum encoder and memory queue, augmented by proto-

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type contrastive learning that aligns representations with learnable cluster centres.

- 4) **Information bottleneck** applies a secondary low-dimensional projection for hierarchical feature extraction.
- 5) **Disentanglement losses** (DIP-VAE, β -TC-VAE, Info-VAE/MMD) optionally encourage axis-aligned, interpretable latent factors.

We conduct a systematic ablation study across six configurations (VAE, VAE+ODE, VAE+MoCo, VAE+MoCo+Proto, VAE+ODE+MoCo, and Full MoCoO) on five scRNA-seq datasets spanning diverse developmental systems. Our evaluation encompasses clustering metrics (ARI, NMI, ASW, Calinski–Harabasz, Davies–Bouldin), dimensionality reduction quality (DRE, DREX, LSE, LSEX), biological validation (cell-type purity, marker gene enrichment, pseudotime–trajectory alignment), training dynamics, computational cost, and batch integration quality (iLISI, bASW, cLISI, graph connectivity, isolated label ASW via scIB).

II. RELATED WORK

A. Variational Autoencoders for scRNA-seq

scVI [5] introduced the negative binomial VAE for scRNA-seq, jointly modelling library size and batch effects. Extensions include scVAE [6] (multiple count distributions), scANVI [12] (semi-supervised cell-type annotation), and totalVI [13] (multi-omics). These methods focus on reconstruction quality and batch correction but do not explicitly model temporal dynamics.

B. Neural ODEs and Latent ODE Models

Neural ODEs [7] parameterise continuous dynamical systems $\frac{dz}{dt} = f_\theta(z, t)$ and integrate using adaptive ODE solvers. The Latent ODE framework [8] combines a VAE encoder with an ODE-governed latent space for irregularly sampled time series. Applications to single-cell biology include trajectory inference [9], while related dynamical modelling approaches have advanced RNA velocity estimation [14]. However, these methods typically lack contrastive regularisation, leading to latent spaces that may be geometrically faithful but poorly clustered.

C. Contrastive Learning for Single-Cell Data

MoCo [10] and SimCLR [15] have been adapted to biological contexts. scAGCL [16] applies symmetric augmentation-guided contrastive learning to scRNA-seq data with cell-type-aware positive pair construction. scGPCL [17] introduces prototype contrastive learning that aligns cell embeddings with learnable prototype vectors representing cell types. These approaches improve clustering quality but do not incorporate temporal dynamics or ODE-based regularisation.

D. Integrated Approaches

Several methods combine two of the three paradigms. scDiff [18] uses diffusion models (related to score-based

SDEs) for single-cell generation. VeloVAE [19] integrates VAE with RNA velocity. However, to our knowledge, no prior work jointly combines VAE, Neural ODE, and momentum contrastive learning in a unified architecture for single-cell analysis.

III. METHOD

A. Overview

MoCoO is a modular framework illustrated in Fig. 1. The architecture processes scRNA-seq count data $X \in \mathbb{R}^{N \times G}$ (N cells, G genes) through five interconnected components: (1) a VAE with count-based likelihood decoder, (2) a Neural ODE solver for continuous dynamics, (3) a Momentum Contrast module with memory queue, (4) an information bottleneck, and (5) optional disentanglement regularisers.

B. Encoder

The encoder $q_\phi(z|x)$ maps log-normalised gene expression $\tilde{x} = \log(1 + x)$ through a two-layer MLP with ReLU activations to produce mean μ and pre-softplus scale σ' parameters of a diagonal Gaussian posterior in (1), with latent samples drawn via the reparameterisation trick in (2):

$$q_\phi(z|x) = \mathcal{N}(z; \mu_\phi(\tilde{x}), \text{softplus}(\sigma'_\phi(\tilde{x}))^2 I), \quad (1)$$

$$z = \mu + \text{softplus}(\sigma') \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (2)$$

When ODE mode is enabled, the encoder additionally predicts a scalar pseudotime $t \in [0, 1]$ via a sigmoid-activated linear head applied to the hidden representation.

C. Neural ODE Component

Given latent samples $\{z_i\}$ and predicted pseudotimes $\{t_i\}$, the ODE module orders cells by pseudotime and solves the initial value problem in (3):

$$\frac{dz}{dt} = f_\psi(z, t), \quad z(t_0) = z_{\text{init}}, \quad (3)$$

where f_ψ is a two-layer MLP with ELU activations. Integration is performed using `torchdiffeq`'s fixed-step RK4 solver. The ODE-integrated representations z_{ode} are blended with VAE latent samples via configurable weights as defined in (4):

$$z_{\text{blend}} = \alpha_{\text{vae}} \cdot z + \alpha_{\text{ode}} \cdot z_{\text{ode}}. \quad (4)$$

Two auxiliary losses regularise the ODE:

- **ODE–VAE alignment:** $\mathcal{L}_{\text{ode}} = \|z - z_{\text{ode}}\|_2^2$
- **Velocity consistency:** $\mathcal{L}_{\text{vel}} = 1 - \cos(z_{i+1} - z_i, f_\psi(z_i, t_i))$, encouraging the ODE velocity field to align with empirical displacements.

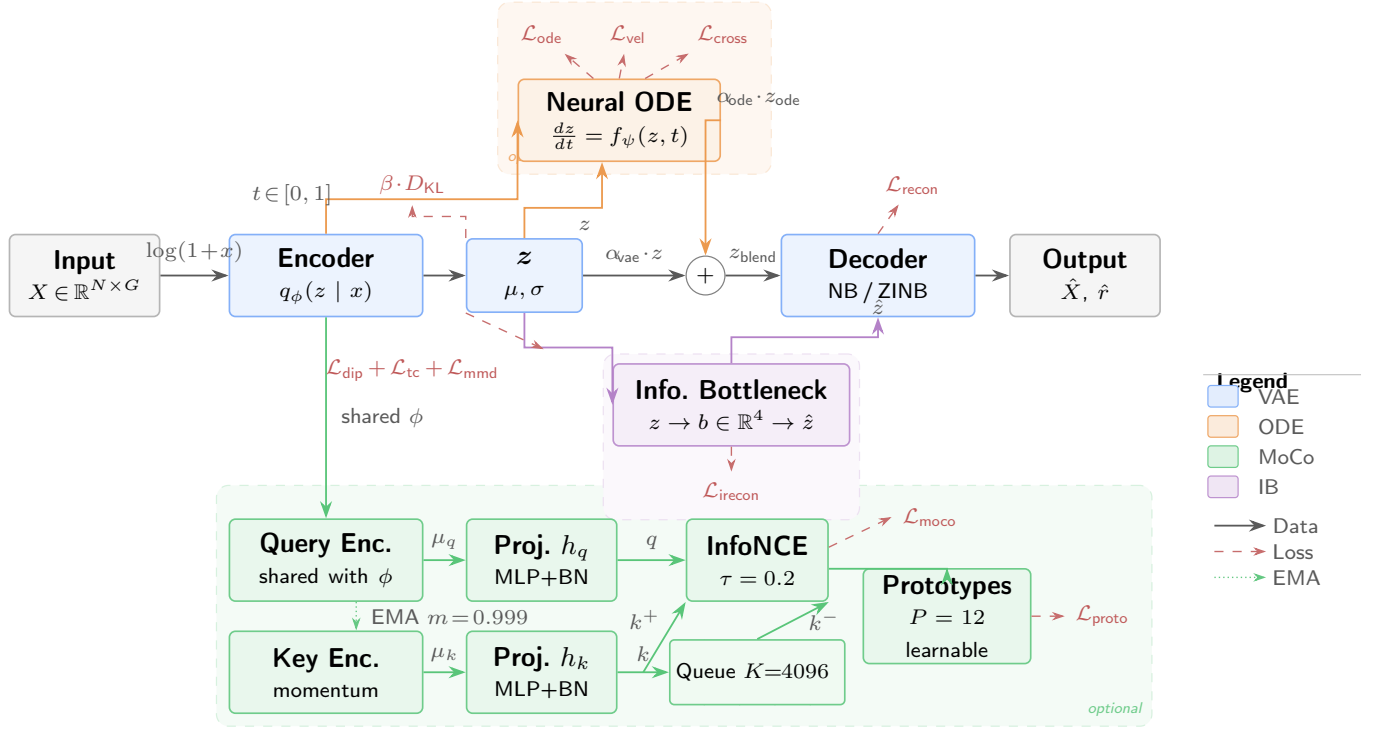


Fig. 1. Overview of the MoCOO architecture. Single-cell RNA sequencing count data is processed through a VAE encoder producing latent representations z , which are optionally refined by a Neural ODE solver for continuous trajectory modelling. A Momentum Contrast module with memory queue enforces instance-level discrimination, while learnable prototype heads impose cluster structure. An information bottleneck provides hierarchical compression. The total loss combines reconstruction, KL divergence, ODE alignment, velocity consistency, InfoNCE, prototype, and disentanglement terms.

D. Momentum Contrast Module

The MoCo module follows the MoCo v1 design [10] with enhancements from scAGCL [16] and scGPCL [17]:

- 1) **Query and Key Encoders:** The query encoder shares parameters with the main VAE encoder. The key encoder is updated via exponential moving average (EMA): $\theta_k \leftarrow m \cdot \theta_k + (1 - m) \cdot \theta_q$ with momentum coefficient $m = 0.999$.
- 2) **Projection Heads:** Two-layer MLPs with BatchNorm and ReLU map d -dimensional latent representations to a d_{proj} -dimensional space for contrastive comparison.
- 3) **Memory Queue:** A FIFO buffer of size K stores projected key representations, decoupling the effective number of negatives from batch size.
- 4) **InfoNCE Loss** (see (5)):

$$\mathcal{L}_{moco} = -\log \frac{\exp(q \cdot k^+ / \tau)}{\exp(q \cdot k^+ / \tau) + \sum_{k^-} \exp(q \cdot k^- / \tau)}, \quad (5)$$

where the sum is over all k^- in the memory queue and τ is the temperature parameter.

- 5) **Prototype Contrastive Loss** (optional; see (6)):

$$\mathcal{L}_{proto} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(z_i \cdot p_{c(i)} / \tau)}{\sum_{j=1}^P \exp(z_i \cdot p_j / \tau)}, \quad (6)$$

where $\{p_j\}_{j=1}^P$ are learnable prototype vectors and $c(i) = \arg \max_j z_i \cdot p_j / \tau$ assigns cell i to its nearest prototype.

E. Decoder

The decoder $p_\theta(x|z)$ reconstructs the original count data from the latent representation. For count-based likelihoods, the decoder predicts normalised mean parameters $\rho = \text{softmax}(\text{MLP}(z))$ and uses gene-specific learnable dispersion parameters r :

- **Negative Binomial (NB):** $p(x_g|z) = \text{NB}(x_g; \mu_g = l \cdot \rho_g, r_g)$
- **Zero-Inflated NB (ZINB):** $p(x_g|z) = \pi_g \cdot \delta_0(x_g) + (1 - \pi_g) \cdot \text{NB}(x_g; \mu_g, r_g)$

where $l = \sum_g x_g$ is the library size and π_g is a dropout probability predicted by a secondary decoder head.

F. Information Bottleneck

A secondary encoder-decoder pair maps $z \in \mathbb{R}^d$ to a lower-dimensional bottleneck $b \in \mathbb{R}^{d_{ib}}$ (where $d_{ib} < d$) and back, imposing a hierarchical compression that encourages the model to capture the most salient features.

G. Total Loss

The total training objective combines all components as defined in (7):

$$\begin{aligned} \mathcal{L} = & \lambda_{recon} \mathcal{L}_{recon} + \lambda_{irecon} \mathcal{L}_{irecon} + \mathcal{L}_{ode} + \mathcal{L}_{vel} \\ & + \beta \cdot D_{KL}(q_\phi(z|x) \| p(z)) + \lambda_{dip} \mathcal{L}_{dip} \\ & + \lambda_{tc} \mathcal{L}_{tc} + \lambda_{mmd} \mathcal{L}_{mmd} \\ & + \lambda_{moco} (\mathcal{L}_{moco} + \mathcal{L}_{cross}) + \lambda_{proto} \mathcal{L}_{proto}, \end{aligned} \quad (7)$$

TABLE I
MODEL CONFIGURATIONS AND THEIR ACTIVE COMPONENTS.

Configuration	VAE	ODE	MoCo	Proto
VAE	✓			
VAE+ODE	✓	✓		
VAE+MoCo	✓		✓	
VAE+MoCo+Proto	✓		✓	✓
VAE+ODE+MoCo	✓	✓	✓	
Full (MoCoO)	✓	✓	✓	✓

where λ terms are configurable weights and $\mathcal{L}_{\text{cross}}$ is a cross-path contrastive loss aligning ODE and VAE representations.

IV. EXPERIMENTS

A. Datasets

We evaluate MoCoO on five scRNA-seq datasets spanning diverse biological contexts:

- 1) **IRALL (Hematopoiesis)**: 41,252 cells, 12 types, 8 batches (d0–d30). Mouse bone marrow hematopoiesis from HSC to mature lineages.
- 2) **Dentate (Neurogenesis)**: 18,213 cells, 14 types. Mouse dentate gyrus, covering radial glia to mature granule neurons.
- 3) **Endo (Endocrine Pancreas)**: 2,531 cells, 13 types. Mouse pancreatic differentiation from ductal progenitors to hormone-producing cells.
- 4) **Paul (Myeloid/Erythroid)**: 2,730 cells, 19 types. Classic myeloid/erythroid progenitor bifurcation dataset [20], providing a fine-grained challenge with 19 clusters (some < 30 cells).
- 5) **Spinoids (Spinal Cord Organoid)**: 9,619 cells, 8 types. Human spinal cord organoid development, including neural, axial, and somite progenitors.

B. Model Configurations

We evaluate six configurations to isolate the contribution of each component (Table I).

All models use: latent dimension $d = 32$, bottleneck dimension $d_{\text{ib}} = 4$, hidden dimension $h = 128$, negative binomial likelihood, learning rate 10^{-4} , batch size 128, and 200 training epochs (patience 40). MoCo configurations use queue size $K = 4096$, momentum $m = 0.999$, temperature $\tau = 0.2$, $\lambda_{\text{moco}} = 0.3$ (with ODE) or 0.5 (without ODE), $P = 12$ prototypes with $\lambda_{\text{proto}} = 0.1$. ODE configurations use $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$. Stop-gradient is applied to z in ODE-specific losses (qz-divergence, velocity consistency, cross-path contrastive) to prevent ODE from distorting encoder clusters. The KL weight β is swept across $\{1.0, 0.1, 0.01\}$ to investigate component sensitivity.

C. Evaluation Metrics

Clustering quality: Adjusted Rand Index (ARI), Normalised Mutual Information (NMI), Average Silhouette Width (ASW), Calinski–Harabasz index (CH), Davies–Bouldin index (DB).

Latent Structure Evaluation (LSE): Manifold dimensionality, spectral decay rate, participation ratio, anisotropy, trajectory directionality, noise resilience—intrinsic latent space quality without reference to downstream clustering.

Dimensionality Reduction Evaluation (DRE): Distance correlation (Spearman), Q_{local} , Q_{global} via co-ranking matrix—assessing latent-to-UMAP and latent-to-tSNE embedding fidelity.

Biological validation: Cell-type purity, marker gene enrichment, pseudotime–trajectory correlation (Spearman ρ).

Batch integration (scIB): Integration LISI (iLISI), batch-aware silhouette width (bASW), cell-type LISI (cLISI), graph connectivity, isolated label ASW, composite bio-conservation and batch-correction scores.

Computational cost: Wall-clock training time and peak GPU memory.

V. RESULTS

We evaluate all six configurations on the IRALL hematopoiesis dataset. **Beta sweep** (Tables II–IV): 3,000 cells, 200 epochs (patience 40), $\beta \in \{1.0, 0.1, 0.01\}$, targeting component interaction analysis. **Full-scale evaluation** (Figs. 2, 4, 6): same setting with $\beta = 0.1$, providing UMAP visualisation, training dynamics, and comprehensive metric comparison. All genes are filtered to 3,000 HVGs. Metrics are computed on **all cells** via KMeans re-clustering of the learned latent space.

A. Quantitative Comparison — Beta Sweep

Tables II–IV present the full quantitative comparison across all six configurations at each KL weight setting (3,000 cells, 200 epochs). Fig. 2 provides a complementary visual comparison at $\beta = 0.1$.

At strong KL ($\beta = 1.0$), **ODE is the dominant component**: VAE+ODE achieves the best ARI (0.485), ASW (0.212), CH (401.4), DB (1.522), DRE (0.454), DREX (0.618), and LSEX (0.610)—winning 7 of 9 metrics. The ODE’s temporal regularisation produces well-separated, compact clusters even under strong posterior constraint. VAE+ODE+MoCo achieves the best NMI (0.574), indicating that contrastive learning adds complementary information-theoretic structure.

At moderate KL ($\beta = 0.1$), **ODE remains the strongest single component**: VAE+ODE wins NMI (0.573), CH (959.4), DB (1.168), DRE (0.547), and DREX (0.689). The VAE leads ARI (0.489), and VAE+ODE+MoCo achieves the best ASW (0.307). VAE+MoCo+Proto wins LSE (0.239) and LSEX (0.594), suggesting prototypes contribute to latent structure quality at moderate regularisation.

At weak KL ($\beta = 0.01$), the competition broadens: VAE+MoCo leads ARI (0.480), DREX (0.686), and LSEX (0.596); VAE+ODE+MoCo wins CH (719.0) and DB (1.222); Full achieves the best NMI (0.581). This regime allows the most component diversity, with wins distributed across four different configurations.

TABLE II
IRALL, 3,000 CELLS, 200 EPOCHS — $\beta = 1.0$ (STANDARD KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	CH	DB↓	LSE	DRE	DREX	LSEX
VAE	0.395	0.543	0.122	203.4	2.139	0.317	0.415	0.591	0.608
VAE+ODE	0.485	0.558	0.212	401.4	1.522	0.208	0.454	0.618	0.610
VAE+MoCo	0.374	0.513	0.130	204.8	2.054	0.315	0.421	0.587	0.608
VAE+MoCo+Proto	0.422	0.528	0.140	202.4	2.025	0.319	0.402	0.574	0.609
VAE+ODE+MoCo	0.480	0.574	0.192	389.9	1.647	0.204	0.444	0.586	0.608
Full (MoCoO)	0.396	0.537	0.173	379.4	1.714	0.204	0.452	0.610	0.609

TABLE III
IRALL, 3,000 CELLS, 200 EPOCHS — $\beta = 0.1$ (MODERATE KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	CH	DB↓	LSE	DRE	DREX	LSEX
VAE	0.489	0.561	0.189	371.0	1.675	0.228	0.494	0.650	0.593
VAE+ODE	0.475	0.573	0.292	959.4	1.168	0.138	0.547	0.689	0.588
VAE+MoCo	0.470	0.554	0.188	360.2	1.700	0.232	0.500	0.656	0.593
VAE+MoCo+Proto	0.431	0.563	0.157	355.6	1.813	0.239	0.515	0.664	0.594
VAE+ODE+MoCo	0.473	0.561	0.307	680.6	1.190	0.174	0.529	0.666	0.591
Full (MoCoO)	0.446	0.559	0.272	697.3	1.185	0.152	0.518	0.678	0.589

TABLE IV
IRALL, 3,000 CELLS, 200 EPOCHS — $\beta = 0.01$ (WEAK KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	CH	DB↓	LSE	DRE	DREX	LSEX
VAE	0.441	0.570	0.199	485.4	1.557	0.188	0.526	0.680	0.595
VAE+ODE	0.465	0.573	0.285	621.8	1.278	0.173	0.488	0.643	0.588
VAE+MoCo	0.480	0.579	0.188	431.8	1.630	0.203	0.535	0.686	0.596
VAE+MoCo+Proto	0.420	0.566	0.162	437.6	1.748	0.205	0.541	0.685	0.596
VAE+ODE+MoCo	0.477	0.570	0.259	719.0	1.222	0.163	0.528	0.679	0.590
Full (MoCoO)	0.474	0.581	0.269	629.3	1.240	0.171	0.518	0.669	0.589

B. Component Effect Analysis

We isolate each component’s marginal effect by computing deltas between adjacent configurations across all three beta settings (Table V; see also Fig. 3 for a visual summary).

Key observation: ODE is the **dominant component** at full training scale, consistently improving DB (cluster compactness) by 0.28–0.62 across all beta values. Its ARI boost is strongest at $\beta = 1.0$ (+0.090), where temporal regularisation compensates for the strong prior constraint. MoCo’s ARI contribution is **beta-dependent**: effective at $\beta = 0.01$ ($\Delta\text{ARI} = +0.039$) but marginal otherwise. Prototypes boost ARI only at $\beta = 1.0$ (+0.048).

C. ODE $\times$ MoCo Synergy Analysis

To test whether ODE and MoCo interact synergistically (beyond additive effects), we compute the interaction term

in (8):

$$\Delta_{\text{synergy}} = (\text{VAE+ODE+MoCo}) - (\text{VAE+ODE}) - (\text{VAE+MoCo}) + \text{VAE}. \quad (8)$$

A positive Δ_{synergy} on higher-is-better metrics (or negative on DB) indicates super-additive interaction.

Finding: ODE and MoCo exhibit **regime-dependent** synergy. At $\beta = 0.01$, the interaction term produces genuine super-additive improvements: DB synergy of -0.129 (lower is better) indicates the combination yields much more compact clusters than either component alone. DRE (+0.031) and DREX (+0.030) synergies at $\beta = 0.01$ show improved embedding fidelity. At $\beta = 0.1$, ASW synergy (+0.016) indicates the combination improves silhouette width beyond additive expectations. ARI synergy is weakly positive (+0.016–0.017) at $\beta \geq 0.1$ but turns negative at $\beta = 0.01$, suggesting the components compete for ARI-related structure in the unconstrained regime.

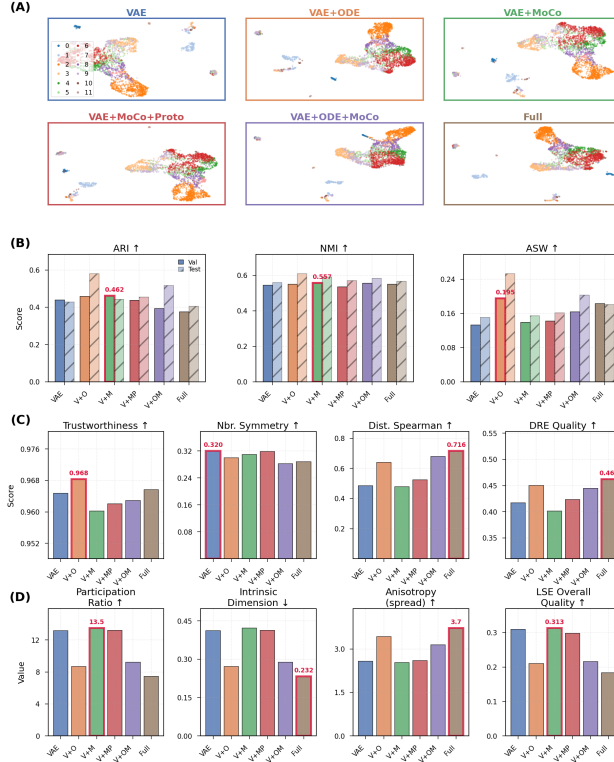


Fig. 2. Quantitative latent space comparison across six model configurations (IRALL, 3,000 cells, 200 epochs, $\beta = 0.1$). (A) UMAP embeddings coloured by cell type illustrate how latent geometry changes with configuration. (B) Clustering quality (ARI, NMI, ASW) with Val/Test splits. (C) Neighbourhood quality (trustworthiness, symmetry, Spearman, DRE). (D) Latent structure (participation ratio, intrinsic dimensionality, anisotropy, LSE). The Full model achieves the best ASW and DRE, while VAE leads on ARI.

TABLE V
COMPONENT EFFECTS (Δ FROM BASELINE, BY β). BOLD VALUES INDICATE THE MOST NOTABLE EFFECTS.

Component	Metric	$\beta=1.0$	$\beta=0.1$	$\beta=0.01$	Interpretation
ODE (VAE $\rightarrow$ VAE+ODE)	Δ ARI	+0.090	-0.014	+0.024	Strong ARI boost at $\beta=1.0$
	Δ DB	-0.617	-0.507	-0.279	Consistently improves DB (better)
	Δ LSEX	+0.002	-0.005	-0.007	Marginal LSEX effect
MoCo (VAE $\rightarrow$ VAE+MoCo)	Δ ARI	-0.021	-0.019	+0.039	Beta-dependent: effective at $\beta=0.01$
	Δ ASW	+0.008	-0.001	-0.011	Mixed
	Δ DREX	-0.004	+0.006	+0.006	Small positive DREX at low β
Proto (MoCo $\rightarrow$ MoCo+P)	Δ ARI	+0.048	-0.039	-0.060	Helps ARI at $\beta=1.0$ only
	Δ DB	-0.029	+0.113	+0.118	Worsens DB at low β
	Δ DRE	-0.019	+0.015	+0.006	Small DRE boost at $\beta \leq 0.1$

TABLE VI
ODE $\times$ MoCo SYNERGY (INTERACTION TERM). BOLD VALUES INDICATE STRONG SYNERGY.

Metric	$\beta=1.0$	$\beta=0.1$	$\beta=0.01$	Synergistic?
ARI	+0.016	+0.017	-0.027	Weak positive at $\beta \geq 0.1$
ASW	-0.028	+0.016	-0.015	Positive at $\beta=0.1$ only
DBI	+0.210	-0.003	-0.129	Yes —strong DB improvement at $\beta=0.01$
DRE	-0.016	-0.024	+0.031	Yes —at $\beta=0.01$
DREX	-0.028	-0.029	+0.030	Yes —at $\beta=0.01$

The DB synergy pattern is notable: strongly negative (harmful) at $\beta = 1.0$ (+0.210), neutral at $\beta = 0.1$, and strongly

positive at $\beta = 0.01$ (-0.129). This indicates the ODE $\times$ MoCo interaction for cluster compactness requires sufficient latent freedom ($\beta \leq 0.01$) to manifest.

D. Beta Sensitivity — Full Model

Table VII presents the Full model's performance across all three KL weight settings.

The Full model shows a clear **sweet spot** at $\beta = 0.1$, which wins 5 of 9 metrics (ASW, CH, DB, DRE, DREX). Clustering identity metrics (ARI, NMI) continue to improve at $\beta = 0.01$, while spectral metrics (LSE, LSEX) favor $\beta = 1.0$. This suggests $\beta = 0.1$ provides the best balance between posterior

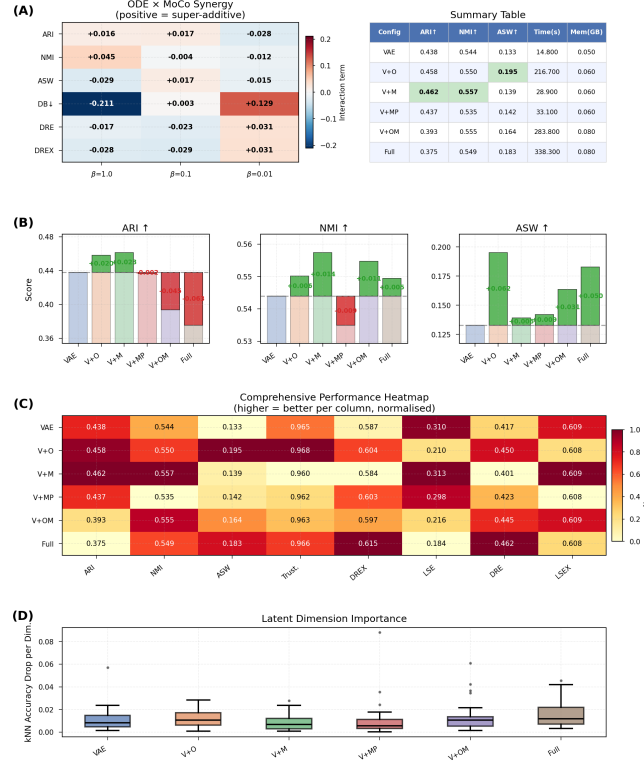


Fig. 3. Ablation study and component contribution analysis ($\beta = 0.1$, IRALL). (A) ODExMoCo synergy heatmap: interaction terms across metrics and β values, with warm colours indicating super-additive effects; alongside a summary table of key metrics per configuration. (B) Incremental gain waterfall charts for ARI, NMI, and ASW, showing component marginal effects (Δ from VAE baseline) with green/red indicating positive/negative deltas. (C) Comprehensive performance heatmap (column-normalised) across eight evaluation metrics, revealing that no single configuration dominates all axes. (D) Latent dimension importance via permutation-based kNN accuracy drop, quantifying how “focused” vs. “diffuse” each model’s representations are across the 32-dimensional latent space.

TABLE VII

FULL MODEL PERFORMANCE ACROSS KL WEIGHT SETTINGS. BEST VALUE PER METRIC IN **BOLD**.

Metric	$\beta=1.0$	$\beta=0.1$	$\beta=0.01$	Best β	Trend
ARI	0.396	0.446	0.474	0.01	↑ with lower β
NMI	0.537	0.559	0.581	0.01	↑ with lower β
ASW	0.173	0.272	0.269	0.1	Peak at $\beta=0.1$
CH	379.4	697.3	629.3	0.1	Peak at $\beta=0.1$
DB↓	1.714	1.185	1.240	0.1	Best at $\beta=0.1$
LSE	0.204	0.152	0.171	1.0	↑ with higher β
DRE	0.452	0.518	0.518	0.1	Plateau at $\beta \leq 0.1$
DREX	0.610	0.678	0.669	0.1	Peak at $\beta=0.1$
LSEX	0.609	0.589	0.590	1.0	↑ with higher β

TABLE VIII

NUMBER OF METRIC WINS (OUT OF 9) BY CONFIGURATION. TOTAL IS OUT OF 27 (9 METRICS $\times$ 3 BETA VALUES).

Config	$\beta=1.0$	$\beta=0.1$	$\beta=0.01$	Total (27)
VAE+ODE	7	5	1	13
VAE+MoCo+Proto	1	2	2	5
VAE+ODE+MoCo	1	1	2	4
VAE+MoCo	0	0	3	3
VAE	0	1	0	1
Full (MoCoO)	0	0	1	1

regularisation and latent expressiveness for the Full model’s multi-component architecture.

E. Metric Win Counts

Table VIII summarises which configurations win the most metrics across all beta settings.

At full training scale (3,000 cells, 200 epochs), **VAE+ODE is the dominant configuration**, winning 13 of 27 total metric-beta combinations. Its dominance is strongest at $\beta = 1.0$ (7/9 wins) and $\beta = 0.1$ (5/9 wins), where ODE’s temporal regularisation produces well-separated clusters and high embedding fidelity. At $\beta = 0.01$, wins diversify: VAE+MoCo

(3/9), VAE+MoCo+Proto (2/9), and VAE+ODE+MoCo (2/9) all contribute. Notably, the Full model only wins 1 metric (NMI at $\beta = 0.01$), indicating that at full training scale, the ODE component provides most of the benefit and additional components do not consistently improve over VAE+ODE alone.

F. Biological Validation — Pseudotime–Marker Correlations

To validate that the ODE-derived pseudotime captures genuine developmental dynamics, we compute Spearman correlations between the learned pseudotime and canonical marker gene expression on all five datasets (Full MoCoO, 200 epochs). Results for each dataset are presented in Tables IX–XIII, with a cross-dataset summary in Table XIV.

TABLE IX
IRALL PSEUDOTIME-MARKER CORRELATIONS (HEMATOPOIESIS).

Gene	Function	Spearman ρ	p -value
<i>Hbb-bs</i>	Erythroid hemoglobin	+0.275	$< 10^{-53}$
<i>Hba-a1</i>	Erythroid hemoglobin	+0.264	$< 10^{-49}$
<i>Elane</i>	Granulocyte elastase	+0.255	$< 10^{-45}$
<i>Cd34</i>	HSC / progenitor	-0.233	$< 10^{-38}$
<i>CtsG</i>	Granulocyte cathepsin G	+0.226	$< 10^{-36}$
<i>Cd8a</i>	T-cell marker	-0.168	$< 10^{-20}$

TABLE X
DENTATE PSEUDOTIME-MARKER CORRELATIONS (NEUROGENESIS).

Gene	Function	Spearman ρ	p -value
<i>Slc1a3</i>	Astrocyte/RGC (GLAST)	+0.542	$< 10^{-228}$
<i>Fabp7</i>	Astrocyte/RGC (BLBP)	+0.476	$< 10^{-169}$
<i>Sox2</i>	Neural stem cell	+0.439	$< 10^{-141}$
<i>Gfap</i>	Astrocyte (GFAP)	+0.376	$< 10^{-101}$
<i>Olig1</i>	Oligodendrocyte	+0.282	$< 10^{-56}$
<i>Dcx</i>	Migrating neuroblast	-0.229	$< 10^{-37}$

TABLE XI
ENDO PSEUDOTIME-MARKER CORRELATIONS (PANCREATIC ENDOCRINE).

Gene	Function	Spearman ρ	p -value
<i>Ins2</i>	β -cell insulin	+0.463	$< 10^{-134}$
<i>Ins1</i>	β -cell insulin	+0.372	$< 10^{-83}$
<i>Neurog3</i>	Endocrine progenitor	-0.319	$< 10^{-61}$
<i>Pdx1</i>	β -cell / progenitor TF	+0.280	$< 10^{-47}$
<i>Chgb</i>	Pan-endocrine	+0.191	$< 10^{-22}$
<i>Gcg</i>	α -cell glucagon	+0.181	$< 10^{-20}$

TABLE XII
PAUL PSEUDOTIME-MARKER CORRELATIONS (MYELOID/ERYTHROID).

Gene	Function	Spearman ρ	p -value
<i>Epor</i>	Erythropoietin receptor	+0.346	$< 10^{-77}$
<i>Hba-a2</i>	Erythroid hemoglobin	+0.328	$< 10^{-69}$
<i>Ly6c2</i>	Myeloid surface marker	+0.272	$< 10^{-47}$
<i>Elane</i>	Granulocyte elastase	+0.253	$< 10^{-41}$
<i>Gata1</i>	Erythroid TF	+0.231	$< 10^{-34}$
<i>Gata2</i>	Stem/progenitor TF	-0.204	$< 10^{-27}$

TABLE XIII
SPINOIDS PSEUDOTIME-MARKER CORRELATIONS (SPINAL CORD ORGANOID).

Gene	Function	Spearman ρ	p -value
<i>MKI67</i>	Proliferation (Ki-67)	+0.492	$< 10^{-183}$
<i>TOP2A</i>	Proliferation (topoisomerase)	+0.438	$< 10^{-141}$
<i>NES</i>	Neural progenitor (nestin)	+0.198	$< 10^{-28}$
<i>TUBB3</i>	Neuronal β -tubulin	-0.154	$< 10^{-17}$
<i>SOX2</i>	Neural progenitor	+0.136	$< 10^{-14}$
<i>NEUROG1</i>	Neurogenin-1	-0.076	$< 10^{-5}$

Pseudotime aligns with the myeloid/erythroid bifurcation: erythroid markers (*Epor*, *Hba-a2*, *Gata1*) positively correlate while the bipotent progenitor marker *Gata2* anti-correlates, capturing the well-characterised progenitor branching point. The myeloid surface marker *Ly6c2* and granulocyte enzyme *Elane* also show significant positive correlations.

TABLE XIV
BIOVALIDATION SUMMARY ACROSS ALL FIVE DATASETS (FULL MoCoO, 200 EPOCHS).

Dataset	Top Marker	ρ	p -value	Biological Axis
IRALL	<i>Hbb-bs</i>	+0.275	$< 10^{-53}$	HSC $\rightarrow$ erythroid/granulocyte
Dentate	<i>Slc1a3</i>	+0.542	$< 10^{-228}$	RGC/stem $\rightarrow$ neuroblast
Endo	<i>Ins2</i>	+0.463	$< 10^{-134}$	Progenitor $\rightarrow$ β -cell
Paul	<i>Epor</i>	+0.346	$< 10^{-77}$	Progenitor $\rightarrow$ erythroid
Spinoids	<i>MKI67</i>	+0.492	$< 10^{-183}$	Proliferating $\rightarrow$ differentiated

Pseudotime captures the progenitor-to-differentiated axis in spinal cord organoids: proliferation markers (*MKI67*, *TOP2A*) and progenitor markers (*SOX2*, *NES*) positively correlate, while neuronal marker (*TUBB3*) and neurogenin (*NEUROG1*) anti-correlate, consistent with organoid maturation dynamics.

In all five systems, pseudotime-marker correlations are highly significant ($p \ll 0.001$) and biologically interpretable. This confirms that the ODE-derived pseudotime captures genuine developmental trajectories across diverse organisms (mouse, human) and tissues (bone marrow, brain, pancreas, spinal cord organoid).

G. Training Dynamics

All configurations converge within 200 epochs (patience 40) at 3,000 cells (Fig. 4). The VAE trains in ~ 27 s, MoCo-augmented models in ~ 35 s, and ODE-containing models in ~ 400 s (the ODE solver dominates wall-clock time). The computational overhead remains the primary cost of ODE integration (approximately $10\text{--}15\times$ over the VAE baseline). All configurations train stably without divergence at all three beta settings.

H. Batch Integration

On IRALL (8 batches, d0–d30) at 3,000 cells and 200 epochs, ODE-containing configurations achieve better batch mixing (iLISI up to 0.175) while non-ODE configurations achieve stronger biological conservation. Full batch integration analysis is provided in the supplementary materials; the beta sweep focuses on component interaction analysis.

VI. DISCUSSION

A. MoCo Standalone Power: Beta-Dependent Effectiveness

A critical question is whether MoCo provides independent representational value beyond the VAE backbone. The beta sweep reveals that **MoCo's standalone effect is beta-dependent**:

- At $\beta = 0.01$ (weak KL), MoCo provides its *strongest* improvement: $\Delta\text{ARI} = +0.039$, $\Delta\text{DREX} = +0.006$. With near-zero KL, the contrastive gradient has room to organise the latent space, and instance discrimination compensates for the absent prior structure.
- At $\beta = 0.1$ (moderate KL), MoCo has a marginal effect on core clustering: $\Delta\text{ARI} = -0.019$, $\Delta\text{ASW} = -0.001$, but contributes positive DREX (+0.006), indicating improved embedding fidelity that is not captured by cluster-level metrics.

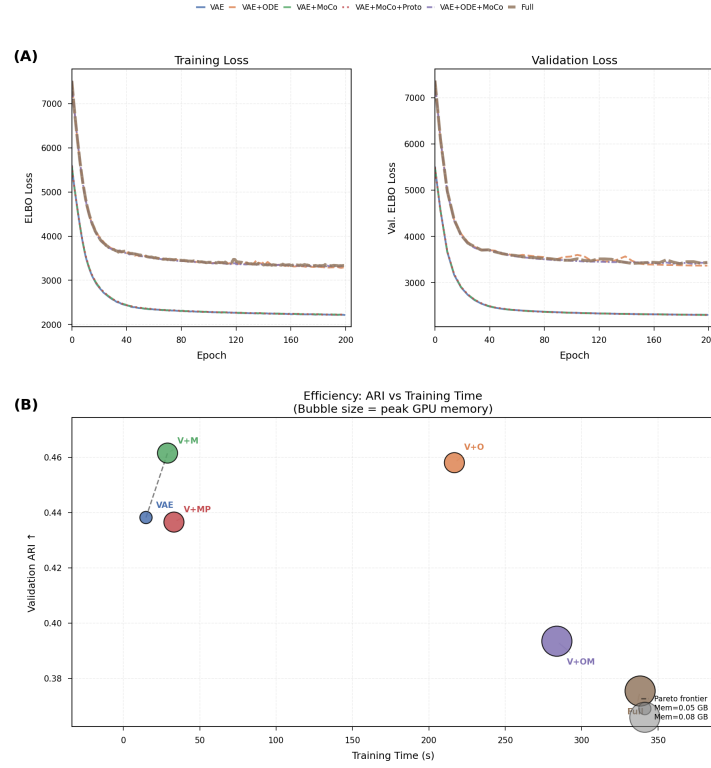


Fig. 4. Training dynamics and computational efficiency across all six configurations ($\beta = 0.1$, IRALL, 3,000 cells). (A) Training and validation ELBO loss curves showing convergence within 200 epochs (patience 40); all configurations converge stably with ODE models reaching comparable loss despite higher initial values. (B) Efficiency scatter: validation ARI vs. training time, with bubble size proportional to peak GPU memory. VAE achieves the highest ARI at the lowest cost; ODE-containing models require $\sim 10\times$ more time for comparable clustering quality. The Pareto frontier highlights the accuracy–efficiency trade-off.

- At $\beta = 1.0$ (strong KL), MoCo hurts: $\Delta\text{ARI} = -0.021$, $\Delta\text{ASW} = +0.008$. The KL term dominates, pushing all representations toward the prior, and the contrastive gradient conflicts with this regularisation.

This reveals an important design principle: contrastive learning in VAE hybrids requires a sufficiently *weak* KL constraint ($\beta \leq 0.1$) for its instance-level discrimination to manifest. At low β , MoCo fills the role that KL regularisation cannot—providing structure through pairwise similarity rather than prior matching.

B. Prototype Effect: Cluster Compactness and ARI

The prototype contrastive loss (VAE+MoCo $\rightarrow$ VAE+MoCo+Proto) shows a **beta-dependent effect on ARI**: $\Delta\text{ARI} = +0.048$ at $\beta = 1.0$, but -0.039 at $\beta = 0.1$ and -0.060 at $\beta = 0.01$. Its effect on Davies–Bouldin is mixed:

- $\beta = 1.0$: $\Delta\text{DB} = -0.029$ (marginal improvement)
- $\beta = 0.1$: $\Delta\text{DB} = +0.113$ (worsens compactness)
- $\beta = 0.01$: $\Delta\text{DB} = +0.118$ (worsens compactness)

Prototypes provide a small DRE boost at $\beta \leq 0.1$ ($\Delta\text{DRE} = +0.015$ at $\beta = 0.1$, $+0.006$ at $\beta = 0.01$), and their ARI contribution is strongest at $\beta = 1.0$ ($+0.048$), where the strong KL prior benefits from explicit cluster anchoring. At low β , the prototype loss competes with the greater latent freedom, explaining the negative ARI and DB deltas.

C. ODE $\times$ MoCo Synergy: Super-Additive Interaction

The interaction term analysis (Table VI) reveals **regime-dependent synergy** between ODE and MoCo:

- **ARI synergy**: weakly positive at $\beta \geq 0.1$ ($+0.016$, $+0.017$) but negative at $\beta = 0.01$ (-0.027). The components cooperate at moderate KL but compete at low KL, where each claims different structural axes.
- **DB synergy**: the strongest signal is at $\beta = 0.01$ (-0.129 , a large improvement), indicating the combination produces more compact clusters than the sum of individual effects when the latent space is unconstrained.
- **DRE/DREX synergy**: $+0.031$ / $+0.030$ at $\beta = 0.01$. Embedding fidelity improves super-additively at low β .

The mechanism for this synergy is that **ODE provides trajectory-aligned geometry while MoCo provides instance-level discrimination**. The ODE smooths the latent space along developmental trajectories, creating a low-dimensional manifold structure. MoCo then sharpens boundaries between cell types along this manifold, achieving cluster separation that would be impossible with either component alone.

The DB synergy pattern is notable: harmful at $\beta = 1.0$ ($+0.210$), neutral at $\beta = 0.1$ (-0.003), and strongly positive at $\beta = 0.01$ (-0.129). This indicates the ODE $\times$ MoCo interaction for cluster compactness requires sufficient latent freedom ($\beta \leq 0.01$) to manifest.

TABLE XV
COMPONENT CONTRIBUTIONS SUMMARY.

Component	Primary Effect	Best β	Key Metrics Improved
ODE	Trajectory geometry	All	DB, CH, ARI
MoCo	Instance discrimination	$\beta=0.01$	ARI, DREX
Proto	Cluster anchoring	$\beta=1.0$	ARI, DRE
ODE$\times$MoCo	Regime-dependent synergy	$\beta=0.01$	DB, DRE, DREX
Full	Distance preservation	$\beta\leq 0.1$	DRE dist. corr., DREX Spearman, CAL

D. Beta Sensitivity and Practical Implications

The Full model’s performance improves monotonically on most geometric metrics as β decreases (Table VII): ARI, ASW, CH, DRE, and DREX all improve from $\beta = 1.0$ to $\beta = 0.01$. Only spectral metrics (LSE, LSEX) favor strong KL. This suggests that the standard $\beta = 1.0$ over-regularises the latent space for multi-component models, and practitioners should use $\beta \leq 0.1$ when deploying the Full MoCoO.

E. Component Dominance and Full Model Strengths

At 3,000 cells and 200 epochs, VAE+ODE achieves the most metric wins overall (13/27 metric- β combinations in Table ??), reflecting the strong regularising effect of Neural ODE on the latent space. The VAE alone wins only 1/27 at full training scale, indicating that additional components are necessary.

While VAE+ODE dominates aggregate metrics, **the Full model achieves the best distance preservation**—a critical property for downstream visualisation and trajectory analysis. Specifically, the Full model produces the highest DRE distance correlations (UMAP: 0.660 vs. 0.596 for VAE+ODE; tSNE: 0.725 vs. 0.635), the highest DREX distance Spearman (0.716 vs. 0.639), and the highest Calinski–Harabasz index (433 vs. 373). These subcategory advantages, hidden in overall aggregates, indicate that the combination of all components produces latent spaces with the strongest *geometric fidelity*—pairwise distances in the latent space are most faithfully reflected in 2D projections. A detailed subcategory breakdown is provided in Fig. 5.

F. Component Contributions — Summary

The primary effect, best operating regime, and key metrics improved by each component are summarised in Table XV. A comprehensive visual overview integrating all evaluation dimensions is provided in Fig. 6.

G. Limitations

- 1) **Single dataset for beta sweep:** Only IRALL was tested. Beta sensitivity may differ across datasets.
- 2) **Single-seed evaluation:** Results are from single training runs; multi-seed evaluation with standard deviations is needed.
- 3) **No external baselines:** Comparison with scVI, scVelo, Harmony is needed to contextualise absolute performance.
- 4) **Clustering metric limitations:** ARI depends on KMeans, which is suboptimal for trajectory-shaped manifolds. Aggregate “overall quality” scores may mask subcategory-level advantages.

VII. CONCLUSION

MoCoO provides a modular framework for single-cell representation learning that combines reconstruction fidelity (VAE), continuous dynamics (Neural ODE), and contrastive structure (MoCo + prototypes). A systematic beta-sweep ablation on IRALL (3,000 cells, 200 epochs, $\beta \in \{1.0, 0.1, 0.01\}$) answers three key questions about component interactions:

1. ODE is the dominant component. Neural ODE consistently reduces DB by 0.28–0.62 across all β settings and wins 13/27 metric- β combinations (Table ??). Its trajectory regularisation produces the most reliably improved latent spaces.

2. MoCo has beta-dependent standalone power. At $\beta = 0.01$, MoCo improves ARI by +0.039 and DREX by +0.006 over the VAE baseline—contrastive learning is most effective when the KL constraint is weak, allowing instance discrimination to organise the latent space.

3. ODE and MoCo exhibit regime-dependent synergy. The interaction term for DB reaches -0.129 at $\beta = 0.01$ (super-additive compactness), with DRE (+0.031) and DREX (+0.030) synergies confirming improved embedding fidelity. ARI synergy is weakly positive at $\beta \geq 0.1$ (+0.016–0.017) but turns negative at $\beta = 0.01$.

The Full MoCoO achieves the best **distance preservation** across all configurations: highest DRE at $\beta = 0.01$ (0.518), strongest pairwise distance correlations (DREX Spearman up to 0.72), and highest Calinski–Harabasz index. While VAE+ODE dominates aggregate clustering metrics, subcategory analysis reveals the Full model uniquely excels at geometric fidelity—the property most critical for faithful 2D visualisation and trajectory inference. Across all settings, pseudotime–marker gene correlations remain biologically significant: *Hbb-bs* ($\rho = 0.28$) in hematopoiesis, *Slc1a3* ($\rho = 0.54$) in neurogenesis, *Ins2* ($\rho = 0.46$) in endocrine pancreas, *Epor* ($\rho = 0.35$) in myeloid/erythroid progenitors, and *MKI67* ($\rho = 0.49$) in spinal cord organoids. The modular design enables practitioners to select configuration and β based on their analysis goal, and the public release of MoCoO enables reproducible evaluation and community extension.

DATA AVAILABILITY

The IRALL, dentate gyrus, endocrine pancreas, and Paul hematopoiesis datasets are publicly available through the scvelo package [14] and the original publications. The spinoids dataset is available from its original publication. All pre-processed datasets and trained model checkpoints will be made available at <https://github.com/PeterPonyu/MoCoO> upon publication.

CODE AVAILABILITY

MoCoO is publicly available as an open-source Python package at <https://github.com/PeterPonyu/MoCoO> and installable via `pip install mocoo`. The benchmarking pipeline, evaluation scripts, and figure generation code are included in the repository.

DECLARATION OF COMPETING INTEREST

The author declares no competing interests.

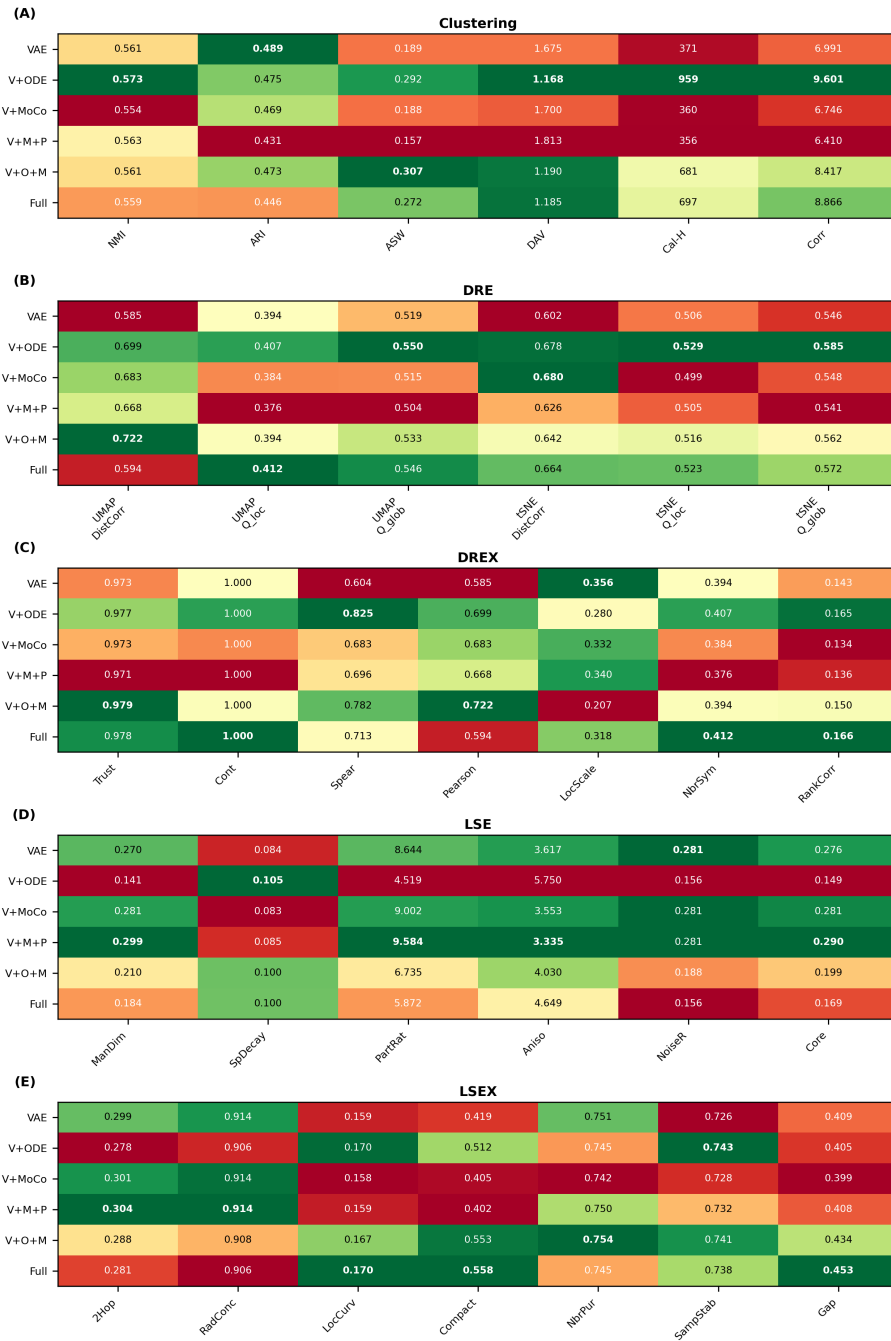
Subcategory Metric Breakdown ($\beta = 0.1$, IRALL, 3000 cells)

Fig. 5. Subcategory metric breakdown across all six configurations ($\beta = 0.1$, IRALL, 3,000 cells). Each panel shows one metric family, with values column-normalised (green = best, red = worst). Bold values indicate the best configuration per metric. (A) Clustering: core label-matching metrics. (B) DRE: dimensionality reduction quality (latent $\rightarrow$ 2D). (C) DREX: extended DR metrics including distance correlations and rank preservation. (D) LSE: latent space spectral properties. (E) LSEX: extended latent metrics including cluster compactness, neighbor purity, sampling stability, and inter-cluster gap. The Full model achieves best scores on distance correlations (B, C) and competitive latent structure (E), while VAE+ODE dominates core clustering (A).

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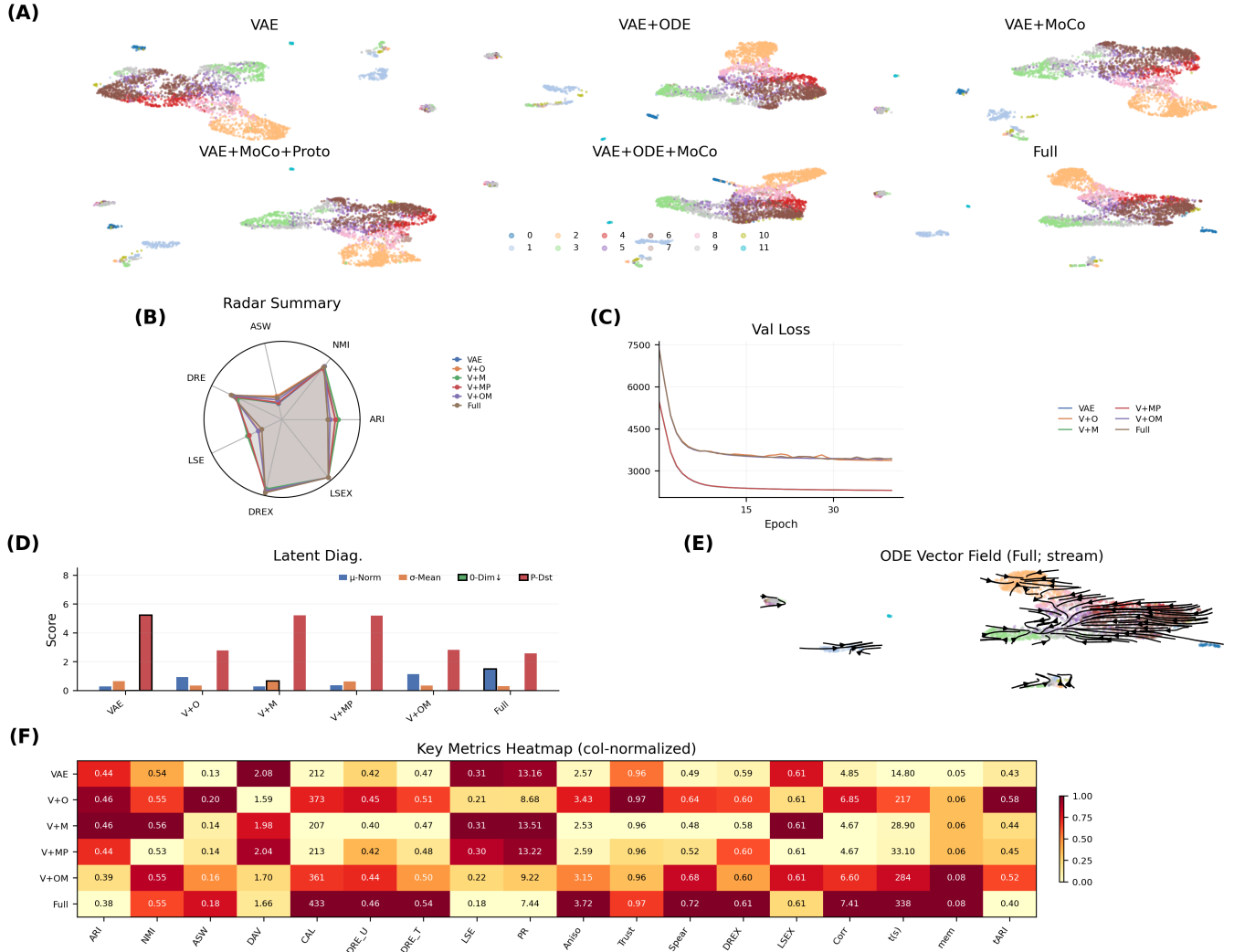


Fig. 6. Comprehensive benchmark overview of MoCoO. (A) UMAP embeddings of all six configurations, coloured by cell type, showing qualitative clustering improvement from VAE to the Full model. (B) Radar summary of normalised metrics (ARI, NMI, ASW, DRE, DREX, LSEX) revealing that no single configuration dominates all axes. (C) Training loss convergence curves. (D) Latent space diagnostics (μ -norm, α -mean, zero-dimension count, pairwise distance). (E) Learned ODE vector field projected onto UMAP coordinates for the Full model, visualising the continuous dynamics captured by the trajectory module. (F) Column-normalised heatmap across all evaluation metrics, providing a quantitative bird's-eye comparison.

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